

A Weighted-Intersection Counterexample for the Kappa Power Question

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2026-07-03

Source and Target Question

The source paper is Braga and Chavez-Dominguez, *On the small scale nonlinear theory of operator spaces*, arXiv:2404.19092 [1]. In Remark 4.7, after computing $\kappa_n(X)$ for several homogeneous Hilbertian operator spaces, the authors ask:

Is $\kappa_n(X)$ always equivalent to a power of n ? That is, does there exist a constant $c(X)$ such that $\kappa_n(X) \simeq n^{c(X)}$?

The answer is negative.

Remark 4.7: The examples for which we have been able to calculate $\kappa_n(X)$ suggest the following two questions for an arbitrary homogeneous Hilbertian operator space X :

- (1) Is the inequality in Proposition 4.2 always an equality? That is, do we have $\kappa_n(X)\kappa_n(X^*) = \sqrt{n}$ for all $n \in \mathbb{N}$?
- (2) Is $\kappa_n(X)$ always equivalent to a power of n ? That is, does there exist a constant $c(X)$ such that $\kappa_n(X) \simeq n^{c(X)}$?

Idea of the Proof

The examples in the source paper give exact powers

$$\kappa_n((R, C)_\theta) = n^{\theta/2}.$$

The class of homogeneous Hilbertian operator spaces is stable under a simple diagonal weighted intersection. If several homogeneous Hilbertian structures on the same Hilbert space are placed in an ℓ_∞ -sum through the diagonal map

$$x \mapsto (\lambda_i x)_i,$$

with $\sup_i \lambda_i = 1$, the scalar norm remains Hilbertian and homogeneity follows coordinatewise. For the special column matrix defining κ_n , the new norm is the supremum of the weighted old κ_n 's.

Choosing weights 2^{-k} and powers $n^{(1-1/k)/2}$ produces the envelope

$$\max \left\{ 1, \sup_{k \geq 2} 2^{-k} n^{(1-1/k)/2} \right\}.$$

Optimizing $2^{-k}n^{-1/(2k)}$ gives a loss of order $\exp(-O(\sqrt{\log n}))$ below $\sqrt{n}$. Thus the sequence has logarithmic exponent tending to $1/2$, but it is not equivalent to $\sqrt{n}$. It therefore cannot be equivalent to any power.

Definitions

Let $H = \ell_2$. We use R and C for the row and column operator space structures on H . For $\theta \in [0, 1]$, $(R, C)_\theta$ denotes the complex interpolation operator space between R and C , over the same underlying Hilbert space H . The source paper proves

$$\kappa_n((R, C)_\theta) = n^{\theta/2} \quad \text{for all } n \in \mathbb{N}.$$

A Weighted Diagonal Intersection Lemma

Lemma 1. *Let $(E_i)_{i \in I}$ be homogeneous Hilbertian operator space structures on the same Hilbert space H , and let $0 < \lambda_i \leq 1$ with $\sup_i \lambda_i = 1$. Define matrix norms on H by*

$$\|[x_{pq}]\|_{M_m(X)} = \sup_{i \in I} \lambda_i \|[x_{pq}]\|_{M_m(E_i)}.$$

Then X is a homogeneous Hilbertian operator space. Moreover, for every n ,

$$\kappa_n(X) = \sup_{i \in I} \lambda_i \kappa_n(E_i).$$

Proof. The norms are realized by the diagonal embedding

$$J : H \longrightarrow \bigoplus_{i \in I}^{\ell_\infty} E_i, \quad Jx = (\lambda_i x)_{i \in I}.$$

Thus Ruan's axioms are inherited from the ℓ_∞ -sum. At matrix level one,

$$\|x\|_X = \sup_i \lambda_i \|x\|_H = \|x\|_H,$$

so X is Hilbertian as a Banach space.

If $T : H \rightarrow H$ is bounded and $m \in \mathbb{N}$, homogeneity of each E_i gives

$$\|T_m([x_{pq}])\|_{M_m(X)} = \sup_i \lambda_i \|T_m([x_{pq}])\|_{M_m(E_i)} \leq \|T\| \sup_i \lambda_i \|[x_{pq}]\|_{M_m(E_i)} = \|T\| \|[x_{pq}]\|_{M_m(X)}.$$

Hence $\|T\|_{\text{cb}: X \rightarrow X} \leq \|T\|$, while the reverse inequality is automatic from level one. So X is homogeneous.

Finally, if $e_1, \dots, e_n$ is orthonormal in H and $b \in M_n(X)$ is the column matrix with entries $b_{j1} = e_j$, then

$$\|b\|_{M_n(X)} = \sup_i \lambda_i \|b\|_{M_n(E_i)} = \sup_i \lambda_i \kappa_n(E_i).$$

This is exactly $\kappa_n(X)$. □

Counterexample

Theorem 1. *There exists an infinite-dimensional homogeneous Hilbertian operator space X such that $\kappa_n(X)$ is not equivalent to n^c for any constant c .*

Proof. Let $E_0 = R$ with $\lambda_0 = 1$. For each $k \geq 2$, let

$$E_k = (R, C)_{1-1/k}, \quad \lambda_k = 2^{-k}.$$

Apply the lemma to obtain a homogeneous Hilbertian operator space X on $H = \ell_2$. Since $\kappa_n(R) = 1$ and, by Corollary 4.3 of [1],

$$\kappa_n(E_k) = n^{(1-1/k)/2},$$

we have

$$\kappa_n(X) = A_n := \max \left\{ 1, \sup_{k \geq 2} 2^{-k} n^{(1-1/k)/2} \right\}.$$

We now show that (A_n) is not equivalent to any power. Put $a = \log 2$ and $L = \log n$. For $k \geq 2$,

$$2^{-k} n^{(1-1/k)/2} = \sqrt{n} \exp \left(-ak - \frac{L}{2k} \right).$$

By the arithmetic-geometric mean inequality,

$$ak + \frac{L}{2k} \geq \sqrt{2aL}.$$

Therefore

$$\frac{A_n}{\sqrt{n}} \leq \max \left\{ n^{-1/2}, \exp(-\sqrt{2a \log n}) \right\} \rightarrow 0.$$

So A_n is not equivalent to $n^{1/2}$.

Conversely, for large n , choose

$$k_n = \left\lfloor \sqrt{\frac{\log n}{2 \log 2}} \right\rfloor \geq 2.$$

Then k_n is within a constant factor of $\sqrt{\log n}$, so there is a universal constant $C > 0$ such that

$$ak_n + \frac{\log n}{2k_n} \leq C\sqrt{\log n}.$$

Thus

$$A_n \geq \sqrt{n} \exp(-C\sqrt{\log n}).$$

It follows that, for every $c < 1/2$,

$$\frac{A_n}{n^c} \geq n^{1/2-c} \exp(-C\sqrt{\log n}) \rightarrow \infty.$$

Since always $1 \leq A_n \leq \sqrt{n}$, equivalence to n^c would force $0 \leq c \leq 1/2$. The preceding two estimates exclude $c = 1/2$ and all $c < 1/2$. Hence no such c exists. $\square$

Verification Notes

The only structural input is that the diagonal weighted intersection is a concrete operator subspace of an ℓ_∞ -sum. Thus the operator-space axioms are not being postulated; they are inherited from a concrete construction. The homogeneity check is coordinatewise and uses exactly the source paper’s definition.

The growth estimate is analytic. The helper script `code/growth_verifier.py` computes the logarithmic finite maximum at representative values of $L = \log n$, reporting $\log(A_n/\sqrt{n})$ and $\log(A_n/n^c)$ for sample $c < 1/2$. It is not used as proof.

Novelty Check

The local registry, solution, attempt, and proof-gap indexes had no prior entry for arXiv:2404.19092. A bounded web search on 2026-07-03 for exact and close phrases, including “ $\kappa_n(X)$ homogeneous Hilbertian operator space”, “ $\kappa_n(X)$ equivalent to a power”, and “ $\kappa_n(X) \kappa_n(X^*) \sqrt{n}$ ”, found the source paper and general background, but no later answer to Remark 4.7.

Limitations

This packet answers only the second question in Remark 4.7. It does not settle the first question, namely whether $\kappa_n(X) \kappa_n(X^*) = \sqrt{n}$ always holds.

References

- [1] Bruno M. Braga and Javier A. Chavez-Dominguez, *On the small scale nonlinear theory of operator spaces*, arXiv:2404.19092, 2024.